SPARK ZERO TO HERO

100+ PAGES

All about Data Engineering!

Spark Tutorial

What is Spark ?

Spark is an **Open Source Unified Computing Engine** with set of libraries for **parallel data processing** on Computer Cluster.

It supports widely used Programming languages such as:

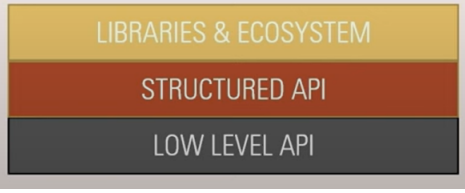
1. Scala
2. Python
3. Java
4. R

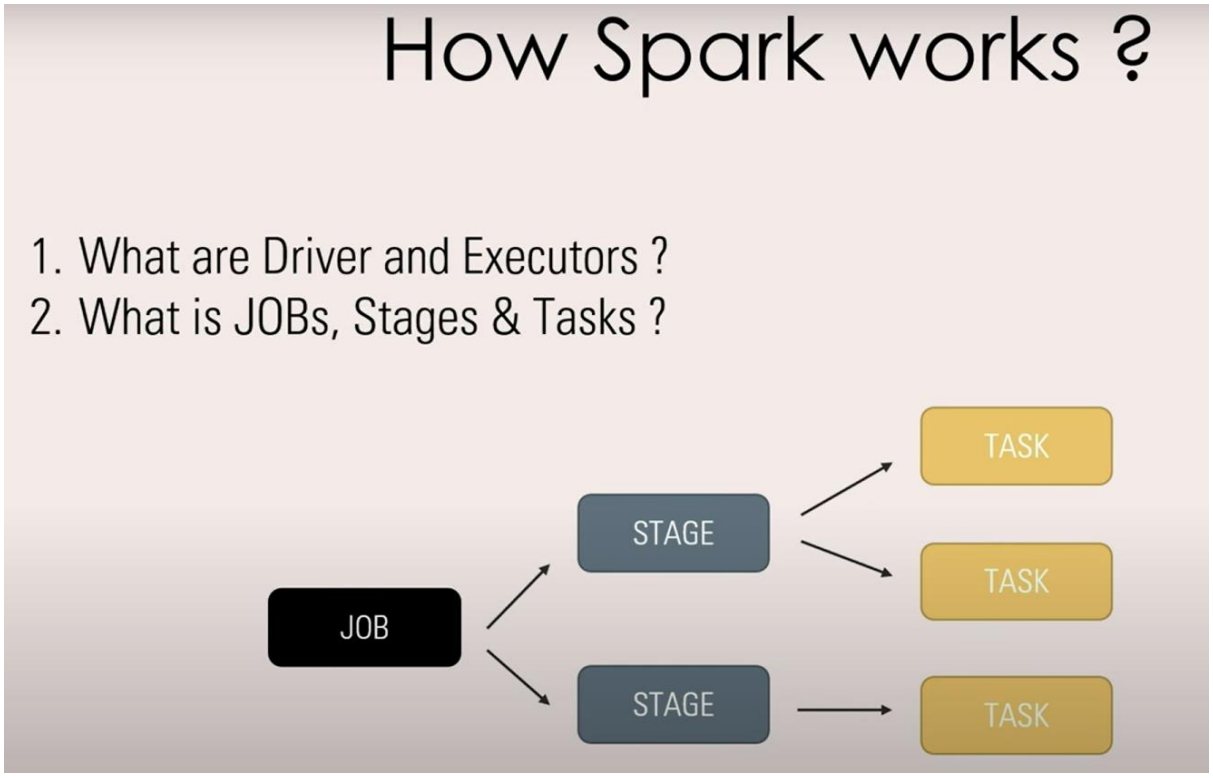
It processes data in memory(RAM) which makes it **100 times faster** than traditional Hadoop Map Reduce.

Spark Components

Following represents Spark Components on High level

1. Low Level API – RDD & Distributed Variables
2. Structured API – DataFrames, Datasets and SQL
3. Libraries and Ecosystem, Structured Streaming and Advanced Analytics





Driver:-

1. **Heart of the Spark Application**
2. **Manages the information and state of executors**
3. **Analyses, distributes and Schedules the work on executors**

Executor:-

1. **Execute the code**
2. **Report the status of execution to driver**

A User assigns a job to driver and driver in turn analyses distribution and breaks down the job into Stages and Tasks. And it assigns it to the executors. Here executors are basically a JVM process which runs in the cluster of machines and it consists of cores.

NOTE:-

1. **Each task can only work on 1 partition of data at a time.**
2. **Tasks can execute in parallel.**